Potential Bias When Using Administrative Data to Measure the Family Income of School-Aged Children

by

Leah R. Clark
U.S. Census Bureau

Renuka Bhaskar
U.S. Census Bureau

CES 25-03 **January 2025**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Sean Wang, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

Abstract

Researchers and practitioners increasingly rely on administrative data sources to measure family income. However, administrative data sources are often incomplete in their coverage of the population, giving rise to potential bias in family income measures, particularly if coverage deficiencies are not well understood. We focus on the school-aged child population, due to its particular import to research and policy, and because of the unique challenges of linking children to family income information. We find that two of the most significant administrative sources of family income information that permit linking of children and parents—IRS Form 1040 and SNAP participation records—usefully complement each other, potentially reducing coverage bias when used together. In a case study considering how best to measure economic disadvantage rates in the public school student population, we demonstrate the sensitivity of family income statistics to assumptions about individuals who do not appear in administrative data sources.

[*] Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7500420: CBDRB-FY2023-CES005-014).

Measures of childhood family income are critical for research and policy. Family income is central to children's short- and long-term outcomes, and certain programs and policies are offered or funded based on the income profile of the child population served (e.g., Title I funding for schools). While annual surveys like the American Community Survey (ACS) document family income for a nationally representative sample of U.S. children, certain research and policy applications require child-level measures at the population scale. For example, the poverty rate of students in an individual school cannot be precisely estimated from a survey sample, even one as large as the ACS. Thus, we turn to administrative data, which—especially when stemming from systems that operate at the state or federal level—can offer consistent measures of family income for the whole population.

However, in order provide consistent measures of family income at the population level, administrative data systems need to include the whole population *and* the information necessary to match individuals across data sources. Existing research suggests linked datasets rarely accomplish that ideal, and, when they fall short, there is risk of producing biased statistics (Bee et al. 2023). Some populations are harder to link across data sources than others, potentially due to missing or erroneous personally identifiable information (PII). For example, Hispanic, non-English-speaking, and non-citizen ACS respondents tend to have lower linkage rates (Bond et al. 2014). Second, there is selection into administrative data systems, since, in the U.S. context, participation is rarely mandatory. For example, filing a federal tax return is optional for very low-income individuals, low-income mothers who file taxes are less likely to claim their children, and unmarried parents are less likely to file than married parents (Gee et al. 2024, Cilke 1998). Taking this evidence together, when it comes to using linked administrative sources of family income in child populations, we expect our income measures to miss certain child subgroups disproportionately—low-income children, in particular. Thus, income measures derived from linked data may be broadly inaccurate if the underrepresentation of low-income children is substantial, and may result in inaccurate statistics about the underrepresented groups.

Do administrative sources of family income, in fact, miss low-income children disproportionately? To answer this, we require a "ground truth" data source that reflects a reasonably representative sample of the child population and already has a measure of family income. We identify two such ground truth sources: (1) the *ACS sample*, which includes all school-aged child respondents to the ACS, and (2) the *K-12 sample*, which includes all public K-12 students from a single state. We further require candidate administrative income data sources that permit the measurement of family income and the identification of children in families. We focus on two major administrative data sources that satisfy these requirements: IRS Form 1040 (tax return) records and Supplemental Nutrition Assistance Program (SNAP) enrollment records. We link each ground truth sample to our candidate income data sources, and document the characteristics—with an emphasis on family income—of children who are or are not found in the candidate sources. Since we have K-12 administrative data for one anonymous state, we opt to restrict our analyses to that state.

Though we use K-12 administrative data as a ground truth sample, it provides an example of another potential issue when relying on administrative measures: the conceptual meaning of a given administrative measure may quietly change over time as programs and policies change. The introduction of universal free lunch in many schools and school districts in the 2010s created new variability in how schools and districts measured students' free and reduced-price lunch eligibility (Downes & Killeen 2024). Although free and reduced-price lunch eligibility has long been used as a proxy for child family income in school accountability tracking, funding formulas, and research, these changes have made it unreliable. Subsequently, public education administrators and researchers are seeking out alternative measures of student family income, with direct certification—an indicator of economic disadvantage drawn from linked administrative data including SNAP—as the best current alternative (Spiegel et al. 2024). Thus, we focus our analyses on the school-aged population and provide an illustrative use case of how different sources and approaches to measuring income with linked administrative data can yield different conclusions about rates of economic disadvantage among public school students.

We proceed with a detailed discussion of our data and methods, then present our results and a concluding discussion.

Data & Method

Conceptual approach

Identifying which individuals do not appear in a candidate administrative data source requires a ground-truth sample that represents the full target population—here, school-aged children. When individuals appear in a ground-truth sample but not a candidate data source there are two general explanations, which we term "unlinkable" and "uncovered." An *unlinkable* record is one which lacks the PII necessary to identify the same individual in another file, or a record that is systematically excluded from the linkage procedure. An *uncovered* record is a linkable record that could potentially appear in the candidate file, but does not. When using administrative records for linked data sources, uncovered records often arise due to selection out of the administrative system, e.g., not filling taxes. Unlinkable or uncovered records may create bias if they are not randomly selected.

The Census Bureau's record linkage process is called the Person Identification Validation System (PVS) and works by assigning Protected Identification Keys (PIKs) to individuals in data sources on the basis of their PII (Wagner & Layne 2014). The reference file for the PVS process is a data file of PII for every social security number (SSN) holder, so this process will include virtually all school-aged children in the ground truth sample who are U.S. citizens. It is not strictly limited to U.S. citizens, however; non-citizen children in our sample who receive SSN's during childhood or as young adults (through subsequent naturalization or work authorization) will also appear in the reference file.

We calculate the proportion of records in the ground truth file that are unlinkable by calculating the proportion without PIKs. We interpret this as an estimate of the proportion of children who have poor quality PII in the ground truth data or are excluded from the PVS reference

file. This approach misses records with PIKs that are, in fact, unlinkable because their record in the candidate administrative data was not correctly assigned their PIK, potentially due to faulty PII in the candidate data source. In the context of this paper, we think this circumstance is rare because the candidate administrative data sources have SSNs and thus achieve near-universal PIK assignment rates.

Ground-truth records with PIKs may be linked to the candidate data source. We calculate the proportion of records in the ground-truth file that are uncovered by calculating the proportion of records that have PIKs but do not appear in the candidate data source. We interpret this as an estimate of the proportion of children whose families select out of the candidate data source or whose families are categorically excluded from the candidate data source.

In addition to documenting linkage and coverage rates, we can directly document the family income profiles of unlinkable and uncovered children, and compare them to those of the linked population. This provides insight into the nature of bias in income measurement that arises due to these excluded subgroups. We also compare linkage and coverage rates across critical subgroups of interest: married and unmarried families, Hispanic and non-Hispanic children, and children enrolled and not enrolled in public school.

Short of achieving a perfect data linkage, the best-case scenario when analyzing a linked dataset is to have an excluded group that is observably similar to the overall population. Such a scenario would support the assumption that records without administrative measures are essentially missing-at-random. This matters because researchers make this assumption—often implicitly—when they restrict analyses to linked records alone.

Selection in candidate data sources

Knowledge about the administrative systems that create the candidate income data sources is important to (a) understand which, if any, children may be categorically excluded from a data source, and (b) create informed priors about which children may not appear due to selection on the part of their parents or caregivers.

To appear in the IRS Form 1040 data, children must be claimed as a dependent by an adult on a tax return.[2] Gee et al. (2024) estimate that 88-97 percent of U.S. children are claimed on tax returns. Families with income above certain thresholds are legally obligated to file a tax return—in 2016, this threshold was $20,700 for a married couple and $13,350 for an unmarried head of household—but families with income below the filing requirement threshold still have considerable incentives to file. If they have any earned income, they are likely to receive a refund and qualify for the Earned Income Tax Credit (EITC) and the Child Tax Credit. Yet nearly one-in-five eligible taxpayers do not claim the EITC (Jones 2014). Families with children are more likely to file taxes, but unmarried parents less so than married parents, and, among tax-filing mothers, child claiming rates are lower at very low incomes (Cilke 1998, Gee et al. 2024). Thus, while the

[2] We exclude tax returns filed by children themselves, favoring a definition of family income that strictly reflects children's adult caretakers' income.

	ACS Sample	K-12 Sample
Target population	School-aged children	Public school students
Years	2012-2017	2012-2014
Baseline income measure	Ratio of income to poverty	Economic disadvantage flag
N	3,772,000	1,670,000
Hispanic	21.8%	21.5%
Not Hispanic	78.2%	78.5%
Married household	67.4%	-
Not married household	32.6%	-
In public school	87.2%	100.0%
Not in public school	12.8%	0.0%

Note: ACS sample N is weighted.

Data sources: ACS 2012-2017, State Department of Education 2012-2014.

Table 1: Description of ground-truth samples

vast majority of children should be claimed on tax returns, we expect those from low-income families to be underrepresented.

Data on SNAP participants, in contrast to IRS Form 1040 data, has a clear categorical exclusion: families not eligible for SNAP benefits. Eligibility is typically determined by documenting household income less than 130% of the federal poverty line, so we can infer that families enrolled in SNAP have income below that threshold, and, depending on the SNAP data system, observe a direct measure of their income. Though SNAP data provides no direct measure of income for families above the threshold, can we infer that those who are not enrolled have income above 130% of the federal poverty line? No, because SNAP take-up is not universal, so some low-income families who are eligible for SNAP are not enrolled (Cunnyngham 2023, Foster et al. 2021).

We primarily position SNAP data as a supplement to IRS data when documenting family income. SNAP benefits increase as income decreases, so a family with zero earned income, who is not required to and might not benefit from filing taxes, might still receive substantial benefits from SNAP. Conversely, high-income families have a legal obligation to file taxes, though they are ineligible for SNAP. Thus, though both data sources reflect selection, the nature of the selection in each might offset the other.

Ground-truth samples and measures of income

For the ACS sample, we use all school-aged child respondents to the ACS from 2012-2017 in our anonymous state of study who are in the age range between kindergarten eligibility (based on the state's rules) and on-time high school completion, approximately ages 5-18. We exclude group quarters respondents. All of our ACS-based results include person weights.

We measure family income in the ACS as the ratio of income-to-poverty for households, where income is measured as household income net of public assistance and SSI. We then assign households to their SNAP eligibility threshold based on their household size, and calculate the ratio of income-to-poverty as income divided by the SNAP eligibility threshold, divided by 1.3 (to scale to poverty rather than SNAP eligibility). We retain imputed values, though imputed values

may themselves be biased due to a missing-at-random assumption in the imputation process (Bollinger et al. 2019, Meyer & Goerge 2011). We discuss this concern in more detail in the results section.

We consider demographic characteristics of interest based on the existing literature on linkage and coverage bias: children's ethnicity, whether children live in a married-parent household, and whether children are enrolled in public school.

For the K-12 sample, we use all students enrolled in public school in grades K-12 in the 2011-12, 2012-13, and 2013-14 school years. We restrict our attention to school years prior to 2014-15 to avoid the potential measurement issues in free/reduced-price lunch eligibility preceded by changes to the National School Lunch Program (Spiegel et al. 2024).

We use an indicator for economic disadvantage as our ground-truth income measure for the public school student population. This binary measure predominantly reflects student eligibility for free/reduced price lunch, though it also factors in other student circumstances, such as homelessness. While not a perfect income proxy, this measure has been shown to capture educational disadvantage in a manner that family income measures do not, so it offers a useful measure for potential selection in the K-12 student population context (Domina et al. 2017). Students are eligible for free lunch if their household income is less than 130% of the federal poverty line and reduced-price lunch if their household income is between 130 and 185% of the federal poverty line. SNAP recipients are categorically eligible for free lunch under the National School Lunch Program's direct certification policy (Gleason 2008).

Table 1 summarizes the two ground truth samples. Hispanic children comprise approximately 22% of both samples. By definition, the K-12 sample is entirely enrolled in public school compared to 87% of the ACS sample. Nearly one-third of school-aged children in the ACS sample reside in households that are not headed by a married couple, but we were unable to construct this attribute from the K-12 data alone.

Deriving income from candidate sources

Our focus is on linkage and coverage rates, and the potential bias introduced by measuring income using linked candidate administrative data when children in a target population are unlinkable or uncovered. We do not endeavor to validate income measures from candidate sources for the linked population, beyond some simple descriptive statistics. There is already a literature comparing survey-reported income to tax-reported income, often from the perspective that tax-reported income is the gold standard (Johnson & Moore 2008, Bollinger et al. 2019). It is hard to conceive of a source of family income data in the U.S. that could be more reliable, aside from the selection concerns central to this paper.

In line with the logic presented above—that IRS and SNAP records offer complementary population coverage—Spiegel et al. (2024) construct a gold standard measure of *school* economic disadvantage from combined IRS/SNAP records. They find that a combined IRS/SNAP approach yields school-level measures virtually identical to those constructed from IRS records alone, suggesting that either the coverage benefits from including SNAP are overstated or they are simply

	All		Sample with PIKs			
	Has PIK	No PIK	Has IRS	No IRS	Has IRS/SNAP	No IRS/SNAP
ACS Sample	92.5	7.5	90.3	9.7	94.5	5.5
Hispanic	87.9	12.1	89.0	11.0	94.5	5.5
Not Hispanic	93.8	6.2	90.7	9.3	94.5	5.5
Married household	93.4	6.6	92.6	7.4	94.6	5.4
Not married household	90.7	9.3	85.7	14.3	94.3	5.7
In public school	92.7	7.3	90.4	9.6	94.8	5.2
Not in public school	91.4	8.6	89.5	10.5	93.0	7.0
Public K-12 Sample	96.5	3.5	87.6	12.4	93.7	6.3
Hispanic	90.9	9.1	85.0	15.0	93.0	7.1
Not Hispanic	98.0	2.0	88.3	11.7	93.9	6.1

Data sources: ACS 2012-2017, State Department of Education 2012-2014, IRS Form 1040 2011-2016, State SNAP 2011-2017.

Table 2: Linkage and coverage rates

not that relevant in the context of school-level measures. The authors argue for the latter, noting that the implicit assumption that IRS-derived economic disadvantage is missing-at-random at the school level appears to be justified. This does not, however, obviate the selection concern for relying on only IRS records when calculating student-level economic disadvantage, as a missing-at-random at the population level assumption is much stronger. We explore this further in the case study that concludes our results section.

SNAP enrollment is itself a qualification for free/reduced-price lunch enrollment, and is therefore a reasonable proxy for student economic disadvantage. In fact, in the absence of reliable free/reduced-price lunch-based economic disadvantage indicators, SNAP enrollment is a highly-reliable proxy for school-level economic disadvantage rates (Spiegel et al. 2024). The potential for deriving continuous measures of income from SNAP administrative data (either through direct income reports or imputation based on benefit amount) is outside of the scope of this paper, but deserves further investigation.

Results

Unlinkable and uncovered children

Before documenting potential bias in income measurement specifically, we establish linkage and coverage rates for our ground-truth samples overall and for key subgroups. Table 2 shows the percentage of the target population that receives PIKs. This can be interpreted as the percentage of the population that is linkable using a routine linkage procedure. Overall, 7.5 percent of the ACS-based school-aged sample does not receive a PIK. Consistent with the Bond et al. (2014) study about PIK receipt in the general ACS sample, Hispanic children and children in non-married households are harder to link. Hispanic children are twice as likely as non-Hispanic children to not be assigned a PIK. Children in non-married households are not assigned PIKs at rates 40 percent higher than children in married households. Being enrolled in public school is associated with slightly improved linkage rates.

In the K-12 sample, just 3.5 percent of students do not receive a PIK, compared to 7.3 percent of public school-enrolled children in the ACS sample. Why the higher linkage rate in the K-12 data? This probably reflects the quality of the PII available in school records. Students and schools have many opportunities over the course of student enrollment to catch errors in records. The ACS, in contrast, works from a single report. This suggests a more limited scope for linkage bias when relying on K-12 records, but the bias is not obliviated. While just 2 percent of non-Hispanic public-school students cannot be linked, 9 percent of Hispanic students do not receive a PIK.

Table 2 also reports the coverage rates for the PIK'd subsamples of the two target populations. 9.7 percent of the school-aged population with PIKs are not claimed as dependents in tax returns. Consistent with Cilke (1998), school-aged children in non-married households have substantially lower coverage rates than those in married households: 14.3 percent are not claimed in 1040 records compared to 7.4 percent of children in married households. Hispanic children and those not in public school are also at elevated risk of not appearing in tax records.

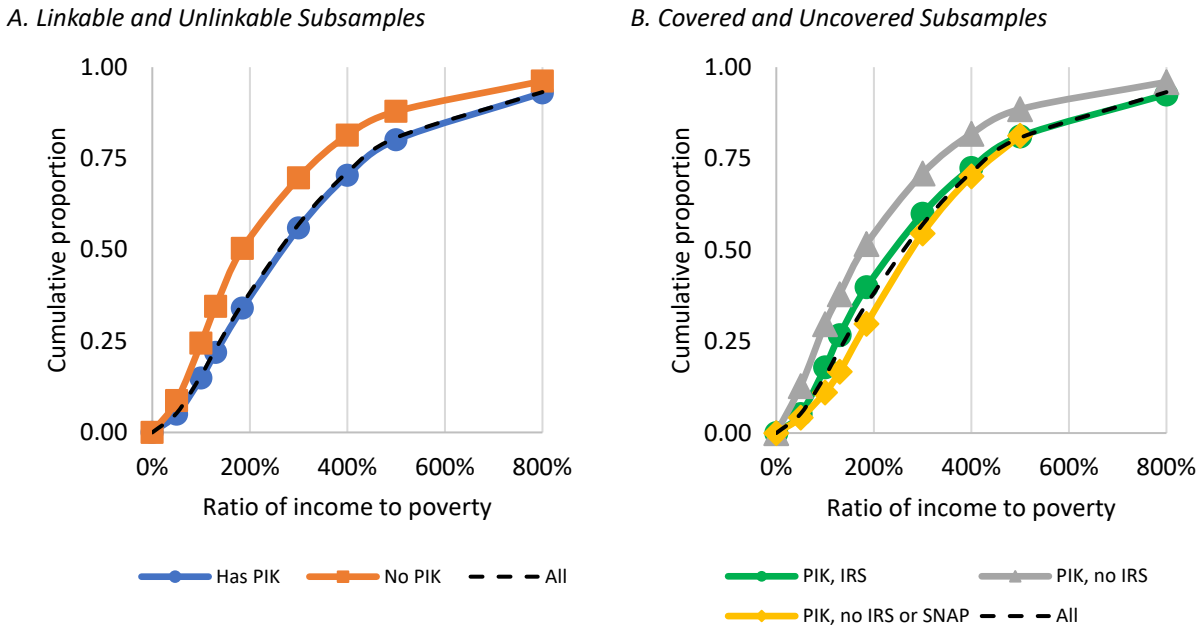
While linkage rates are higher in the K-12 sample, coverage rates are lower. 12.4 percent of K-12 students are not claimed as dependents in the IRS data. This may reflect the fact that low-income students are more likely to attend public school, but less likely to have parents who file taxes. Among public K-12 students, 15.0 percent of Hispanic students are not claimed on tax returns compared to 11.7 percent of non-Hispanic students.

Supplementing SNAP records with IRS records shows promise for reducing bias in terms of evening out the observable characteristics of covered and uncovered populations, as shown in the last two columns of Table 2. Just 6 percent of both the ACS and K-12 populations with PIKs are missing from both IRS and SNAP data sources. The largest documented disparity—that between school-aged children in married and unmarried households—virtually disappears. SNAP records offer improved coverage of children in single-parent families, in particular. The one disparity that does not decrease is that between children enrolled in public school and those who are not. That SNAP data improves coverage for public-school enrolled children, in particular, is consistent with the fact that low-income children disproportionately attend public school.

While the combined SNAP/IRS data source yields identical coverage for Hispanic and non-Hispanic school-aged children in the ACS, a small disparity persists in the K-12 sample. 7.1 percent of Hispanic students are not covered in the combined data compared to 6.1 percent of non-Hispanic students. However, this represents a sizeable improvement in coverage and a much smaller disparity than arises when using IRS records alone.

The family incomes of unlinkable and uncovered children

We now turn to our primary question: do candidate administrative data sources on family income miss low-income children disproportionately? Relying on measures of family income from our ground-truth data sources, we directly document the income profiles of unlinked and uncovered children.



Data sources: ACS 2012-2017, IRS Form 1040 2011-2016, State SNAP 2011-2017.

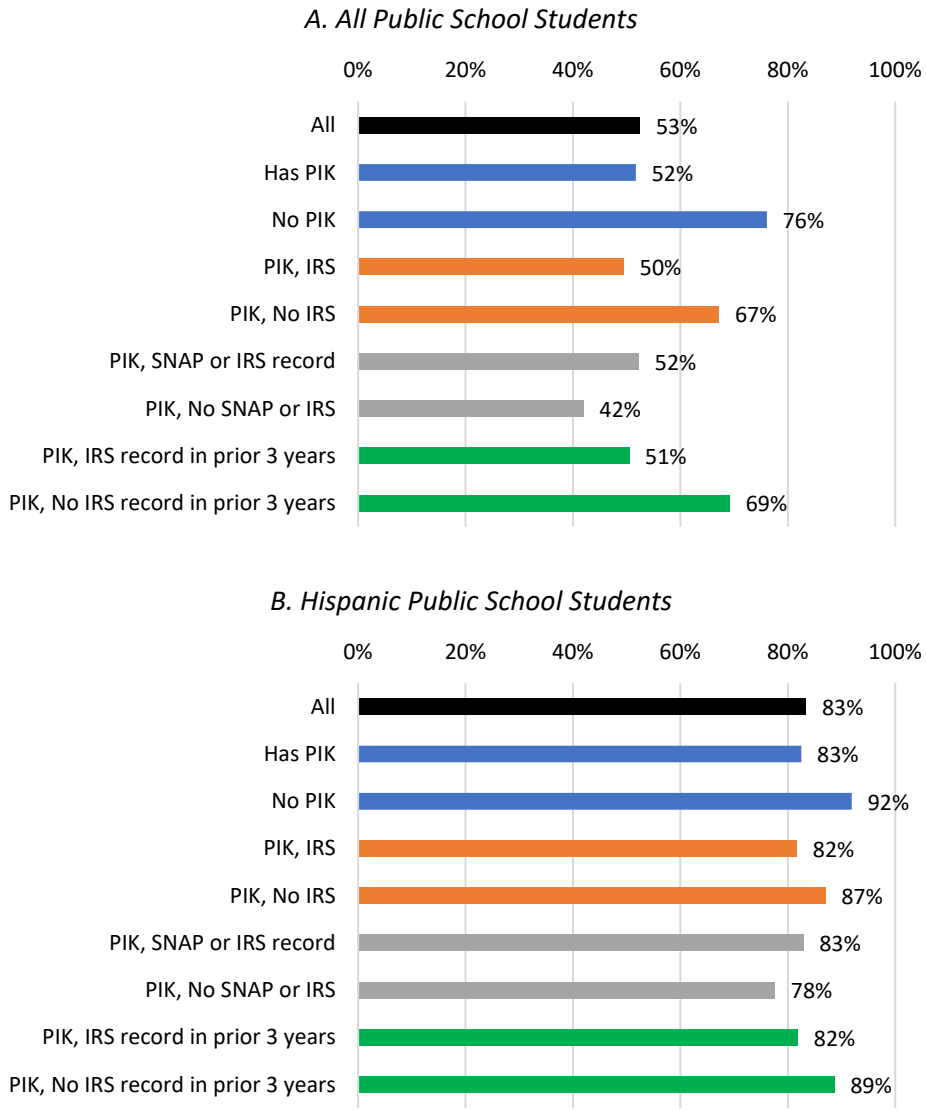
Figure 1: Ratio of income to poverty distribution by linkage and coverage status

For the ACS sample, we plot the cumulative distribution of family income—measured as the ratio of income to the federal poverty line—separately for the linkable (has PIK) and unlinkable (no PIK) subsamples in Panel A of Figure 1. Unlinkable school-aged children have lower family income than linkable children, with the cumulative distribution for children without PIKs lying to the left of that for children with PIKs. For example, 24 percent of unlinkable children have family income below the poverty line, while 15 percent of linkable children do. This indicates that when studies exclude unlinkable children, they exclude a disproportionately low-income group.

Panel B of Figure 1 assesses potential bias in family income measures within the linkable population due to incomplete coverage from administrative data sources. Not surprisingly, uncovered children are generally lower income than covered children. Among school-aged children with PIKs, 30 percent of those who do not appear in IRS records have family income below the poverty line, in contrast to 18 percent of those who do. This is consistent with the expectation that low-income families are less likely to file taxes than higher-income families.

Does supplementing IRS records with SNAP records show promise for reducing coverage bias in family income? To examine this, we further plot the family income distribution of linked school-aged children who appear in neither data source. This subgroup has an income distribution similar to, though actually slightly higher than, the general population. This suggests that SNAP records are effective in identifying low-income children who are not claimed on tax returns, in addition to improving coverage rates overall.

While this result also suggests that children who are not claimed on tax returns and who do not receive SNAP benefits are not typically low-income—but rather have incomes typical of



Note: Panel A includes all students in the K-12 sample, but Panel B restricts the sample to Hispanic students.
Data sources: State Department of Education 2012-2014, IRS Form 1040 2009-2013, State SNAP 2011-2014.

Figure 2: Percent of Public School Students Economically Disadvantaged

the broader population—there is an important caveat to note here: we retain imputed values of ACS-derived income, which might create the appearance of missing-at-random income for households that, in fact, did not report their income and tend to be lower income, on average (Bollinger et al. 2019). Excluding imputed values would likely shift down the income distributions of uncovered populations, potentially revealing that children linked to neither IRS nor SNAP records are lower-income, on average, than the broader population. Fortunately, this concern does not arise in the K-12 sample, which reveals similar patterns to those documented above.

We extend this analysis to the K-12 sample by looking at economic disadvantage rates for students by whether or not they receive a PIK, and, among those with a PIK, whether or not they appear in IRS records or the combined IRS/SNAP records. Panel A of Figure 2 reveals that students

without PIKs have high rates of economic disadvantage, as do those with PIKs who do not show up in IRS records.

Consistent with the findings for the ACS sample, the combined IRS/SNAP approach appears to capture many of the economically disadvantaged students whose families do not file taxes. 67 percent of linkable students uncovered by IRS records are economically disadvantaged, in contrast to just 42 percent of linkable students uncovered by the combined IRS/SNAP data. As with the ACS analysis, students uncovered by the combined data are higher income, on average, than students generally. We also consider whether using multiple years of IRS data—which improves the coverage rate to 94%, in line with the combined IRS/SNAP approach—does anything to reduce income bias, but we find that it does not. Students who are consistently not in IRS data are disproportionately economically disadvantaged, but SNAP data capture many of these students.

We have shown linkage and coverage rates for Hispanic children, but we have not investigated whether income bias might differ for the Hispanic subgroup in our data sources. Thus, we repeat this analysis for Hispanic public-school students in Panel B of Figure 2. Hispanic students have high rates of economic disadvantage, and the differences between linkable and unlinkable or covered and uncovered students are compressed, but follow the same pattern as discussed for the full population. For Hispanic students, using multiple years of IRS data improves coverage to 91 percent (compared to 85 percent for a single year of IRS data) but does not reduce bias, while using the combined IRS/SNAP approach improves coverage to 93 percent and produces a covered sample with the economic disadvantage rate of the overall Hispanic population.

Case study: Measuring student economic disadvantage

We have shown that there is a risk of bias when measuring family income for the school-aged child population using linked candidate administrative data sources. Yet whether bias actually impacts conclusions depends on the specific application, as does the best approach for handling records with missing income. To illustrate this, we compare time trends in different measures of K-12 student economic disadvantage for the anonymous state of study. Our goal is to come as close as we can to replicating the patterns shown by the economic disadvantage indicator in our ground truth sample with an alternative measure derived from our candidate administrative data sources. In trying to construct an alternative measure for this variable, note that this task is eased by the fact that Economic Disadvantage simply categorizes students as relatively low-income or not; we are not trying to match a continuous measure of income which is much more challenging.

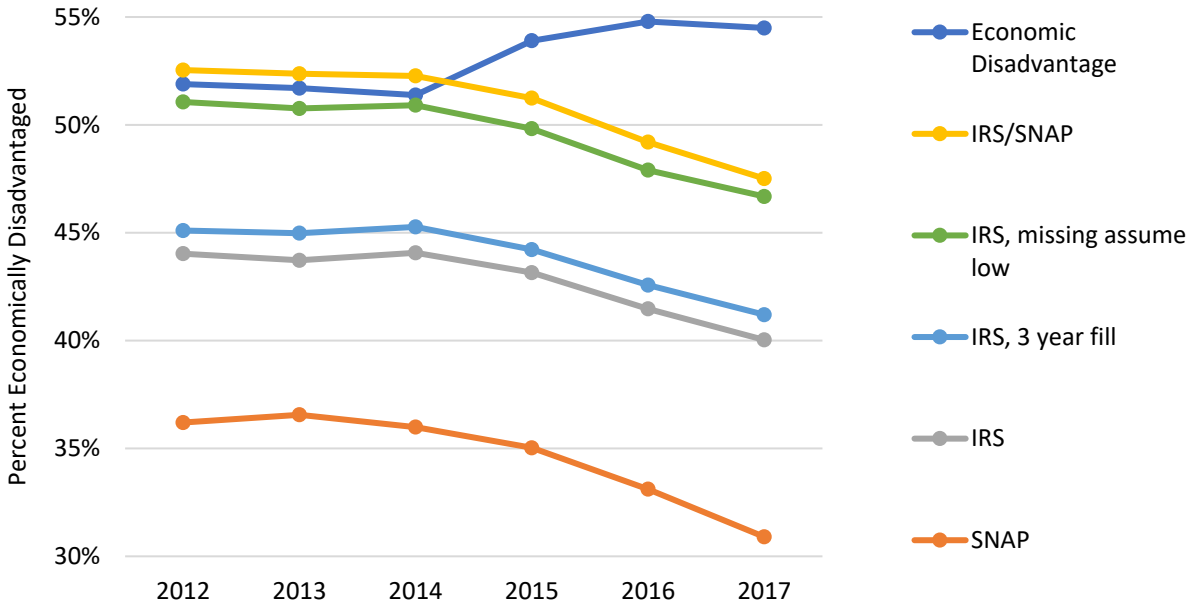
Figure 3 plots the state-wide average over time of the Economic Disadvantage variable against the following indicators constructed from the linked IRS and SNAP data:

1. *IRS.* This variable designates students with PIKs who have IRS-reported family income below 185 percent of the federal poverty line (the threshold for reduced-price lunch) as economically disadvantaged. Students—with or without PIKs—who lack IRS-reported family income are excluded from the analysis. This constitutes an assumption that these students' family incomes are missing at random.

2. *SNAP*. Students with PIKs who are enrolled in SNAP are deemed economically disadvantaged. Students with PIKs who are not enrolled in SNAP are assumed not economically disadvantaged. Students without PIKs are excluded from the analysis (i.e., assumed missing at random). Since SNAP eligibility is 130 percent of the federal poverty line, this measure inherently misses students traditionally designated as economically disadvantaged, and, as a proxy of traditional economic disadvantage, exhibits substantial downward bias. We retain it here because it is conceptually similar to direct certification-based measures which are available in many administrative K-12 sources, and have been shown to be reliable for capturing cross-school variation in economic disadvantage (Spiegel et al. 2024).
3. *IRS/SNAP*. This variable designates all students with PIKs who are enrolled in SNAP as economically disadvantaged. Students with PIKs not enrolled in SNAP are assumed to *not* be economically disadvantaged unless they have IRS-reported family income below 185 percent of the federal poverty line (the threshold for reduced-price lunch). Students without PIKs are excluded from the analysis (i.e., assumed missing at random).
4. *IRS, 3 year fill.* This variable builds from the *IRS* variable by additionally designating as economically disadvantaged students who did not appear in the tax data in the current year but did in the prior year (or two years prior) and had family income below 185 percent of the federal poverty line in that year.
5. *IRS, missing assume low*. This variable alters the *IRS* variable by assuming students who do not appear in tax return data—including those without PIKs—are low income. Since we know (from Figure 2) that more than two-thirds of these students are economically disadvantaged, this seems a reasonable assumption to consider.

Figure 3 confirms several concerns. First, there is a clear change in the Economic Disadvantage variable where its trend diverges from the IRS/SNAP-based variables starting in 2015. This is likely because changes in the National School Lunch Program unveiled nation-wide in the 2014-15 school year altered the use and meaning of this indicator. This illustrates how variables that might seem similar over time in administrative data can quietly change as the administrative concept they reflect change. Second, SNAP alone, as expected, underestimates the target economic disadvantage threshold, though this does not undermine its proven usefulness. It simply highlights how a measurement approach might be ideal in one setting even as it misses the mark in another.[3] Third, relying on tax-derived income as the sole information source with minimal added assumptions—see the IRS-alone and 3-year-fill lines—underestimates the economically disadvantaged population by approximately 7-8 percentage points. This follows from our earlier findings that students without IRS records are disproportionately low-income.

[3] Specifically, we expect SNAP, which targets 130% of the federal poverty line and below, to underestimate Economic Disadvantage, which targets 185% of poverty and below. Yet Spiegel et al. (2024) show that SNAP does a very good job capturing the same ranking of schools as a measure that targets 185% of poverty.



Data sources: State Department of Education 2012-2017, IRS Form 1040 2009-2016, State SNAP 2011-2017.

Figure 3: Population-wide measures of economic disadvantage among public school K-12 students

Two measures track Economic Disadvantage closely prior to 2015: the combined IRS/SNAP measure and the IRS-based measure that assumes students missing IRS data are universally low-income. Note, however, that we handle unlinkable and uncovered students differently in these two measures. In the latter measure, we designate students without PIKs or IRS income as low-income, knowingly overstating economic disadvantage: our analyses suggest that roughly one-third of unlinkable and uncovered students are not disadvantaged. Conversely, our assumptions behind the IRS/SNAP measure understate economic disadvantage: we designate students with PIKs but without IRS/SNAP records as economically advantaged, when we know 42 percent are not; while we designate students without PIKs as missing-at-random, when we know, in fact, three-quarters of them are economically disadvantaged. If we constructed more neutral assumptions on both measures, the IRS/SNAP measure would shift up and the IRS-only measure down, both away from the Economic Disadvantage measure. Since we are confident in our designation of SNAP recipients as economically disadvantaged and know the inclusion of SNAP records improves coverage over relying on IRS records alone, we are inclined to believe that our "ground truth" Economic Disadvantage measure is slightly underestimated in the state of study prior to 2015.

While we cannot be completely certain of the true level of economic disadvantage in the state, the wide range of values that are derived from our administrative measures under different assumptions is notable. Focusing on the measures that target 185% of the federal poverty line and below, estimated rates of economic disadvantage range from 44-53 percent in 2012 to 40-48

percent in 2017, with the aforementioned caveat that the upper bound on the IRS/SNAP measure under different reasonable assumptions may be be a couple points higher.

Discussion

We provide an overview of potential biases in family income measures that arise when relying on linked administrative income measures, even when administrative sources are high quality and the overwhelming majority of source records are linked. We focus on school-aged children because of their central importance to policy. We show that administrative data sources with complementary weaknesses can work together to improve coverage and measurement. Specifically, the combined use of SNAP and IRS records in characterizing the family income of the child population is promising.

Our study considers only a single state and focuses on a time period where child poverty was stable and then declining. Other states and time periods warrant attention, especially considering variation in SNAP implementation and take-up across states and over time (Foster et al. 2021). In terms of other age groups, our findings may generalize to the population of young children (under age 5), but that group may be harder to link across data sources as their appearance in administrative records may be lagged (Bernard et al. 2024). Measuring income for working-age adults is easier, in the sense that the parent-child linkage is not necessary, so the set of potential data sources is broader (e.g., unemployment insurance records, IRS Form W2). Measuring income for retirement-age adults comes with a different set of challenges, since tax records may not accurately capture their true income. SNAP records may be useful across these populations in identifying low-income individuals who are less engaged in formal employment.

What should researchers and practitioners do about children whose income they cannot measure when using an administrative data source? Meyer & Goerge (2011) suggest using inverse probability weighting to account for missing PIKs. More generally, one can use inverse probability weighting to model the probability any ground-truth record appears in a candidate administrative data source, and then use the inverse probability to upweight the sorts of records that may be underrepresented. The value of this approach depends on the richness of the variables available in the ground truth dataset. Bollinger et al. (2017) find a copula-based approach outperforms the inverse probability weighting approach in handling survey non-response for income. Our limited analyses do not test these approaches, but they warrant deeper examination for handling missingness in candidate administrative data sources, in particular.

When it comes to coarse measures—like the Economic Disadvantage indicator in the K-12 sample—in a context with a high linkage rate, an aggressive assumption—that all students without observed income are economically disadvantaged—can wind up being accurate in an aggregate sense. But such assumptions should made with contextual or analytical support. When it comes to handling missing values of a detailed income measure, an extreme assumption—such that all children with missing income have *zero* family income—appears less reasonable than a missing-at-random assumption, though we show that, too, benefits from the use of complementary administrative sources.

References

Bee, A., Mitchell, J., Mittag, N., Rothbaum, J., Sanders, C., Schmidt, L., & Unrath, M. (2023). National Experimental Wellbring Statistics. *SEHSD Working Paper Series*, No. 2023-02.

Bernard, J., Drotning, K., & Genadek, K. R. (2024). Where Are Your Parents? Exploring Potential Bias in Administrative Records on Children. *CES Working Paper Series*, No. CES 24-18.

Bollinger, C. R., Hirsch, B. T., Hokayem, C. M. & Ziliak, J. P. (2019). Trouble in the Tails? What We Know about Earnings Nonresponse 30 Years after Lillard, Smith, and Welch. *Journal of Political Economy*, 127(5), pp. 2143-85.

Bond, B., Brown, J. D., Luque, A. & O'Hara, A. (2014). The Nature of the Bias When Studying Only Linkable Person Records: Evidence from the American Community Survey. *CARRA Working Paper Series*, No. 2014-08.

Cilke, J. (1998). A Profile of Non-Filers. *OTA Paper*, No. 78.

Cunnyngham, K. (2023). Reaching Those in Need: Estimates of State Supplemental Nutrition Assisstance Program Participation Rates in 2020. *United States Department of Agriculture*.

Domina T., Pharris-Ciurej N., Penner A. M., Penner E. K., Brummet Q., Porter S. R., Sanabria T. (2018). Is free and reduced-price lunch a valid measure of educational disadvantage? *Educational Researcher*, 47(9), 539–555.

Downes, T. & Killeen, K. (2024). What Comes After Lunch?: Alternative Measures of Economic and Social Disadvantage and their Implications. Charlotte, NC: Information Age Publishing.

Foster, T. B., Knop, B., Bhaskar, R. (2021). Demographic SNAPshot: Not Everyone Eligible for Food Assistance Program Receives Benefits. *U.S. Census Bureau America Counts Story*.

Gee, G., Goldin, J., Gray-Hancuch, J., Lurie, I., & Vohra, V. (2024) The Claiming of Children on U.S. Tax Returns. *NBER Working Paper Series*, No. 33277.

Gleason, P. (2008). Direct Certification in the National School Lunch Program Expands Access for Children. *Journal of Policy Analysis and Management,* 27(1), pp. 82-103.

Johnson, B. W., & Moore, K. (2008). Differences in Income Estimates Derived from Survey and Tax Data. *SOI Paper Series.*

Jones, M. R. (2014). Changes in EITC Eligibility and Participation, 2005–2009. *CARRA Working Paper Series*, No. 2014-04.

Meyer, B. D., & Goerge, R. M. (2011). Errors in Survey Reporting and Imputation and Their Effects on Estimates of Food Stamp Program Participation. *CES Working Paper Series*, No, 11-14.

Spiegel, M., Clark, L. R., Domina, T., Radsky, V., Yoo, P. Y., & Penner, A. (2024). Measuring School Economic Disadvantage. *Educational Evaluation and Policy Analysis*.

Wagner, D., & Layne, M. (2014). The Person Identification Validation System (PVS): Applying the Center for Administrative Records Research and Applications' (CARRA) Record Linkage Software. *CARRA Working Paper Series,* No, 2014-01.